



Subrata Golui

Visiting Assistant Professor

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Research Profile

I am a researcher in applied probability, stochastic control, Markov decision processes, and stochastic games. My work develops mathematically rigorous and computationally implementable decision models for systems affected by rare high-cost events, partial observation, and multiple time scales. I completed my Ph.D. in Mathematics at the Indian Institute of Technology Guwahati on 4 July 2023 under the supervision of Dr. Chandan Pal. My doctoral thesis was entitled *Risk-sensitive Stochastic Control and Games*.

Primary Research Areas	Risk-sensitive stochastic control; zero-sum and nonzero-sum stochastic games; discrete- and continuous-time Markov decision processes; piecewise deterministic Markov decision processes; singular perturbations; partially observable control; stochastic differential games
Current Position	Visiting Assistant Professor, Department of Mathematics, IIT Dharwad, India
Research Output	Eleven published or accepted journal articles and three manuscripts under review
Languages	Bengali, English, and Hindi

Selected Research Contributions and Independence

- Developed risk-sensitive stochastic control and game models under discounted, finite-horizon, first-passage, average-cost, and ergodic criteria, with optimal policies, saddle-point equilibria, and stationary Nash equilibria.
- Established dynamic-programming, Hamilton–Jacobi–Bellman, Hamilton–Jacobi–Isaacs, and nonlinear eigenvalue characterizations on countable and general state spaces.
- Developed a controlled production–inventory model and analysed implementable value- and policy-iteration algorithms.
- Extended this programme during postdoctoral research to PDMDPs, singular perturbations, partially observed models, stochastic differential games, and numerical validation. A single-author paper on risk-sensitive PDMDP games provides direct evidence of research independence.
- I am the first or sole author of seven of my eleven published or accepted articles and of all three manuscripts currently under review.

Expertise and Skills

Theoretical Methods	Dynamic programming; HJB/HJI equations; nonlinear eigenvalue methods; Lyapunov techniques; equilibrium analysis; singular perturbation methods
Computational Methods	Value iteration, policy iteration, eigenvalue computation, Monte Carlo simulation, Markov Chain Monte Carlo (MCMC), Gillespie simulation, and numerical implementation of stochastic processes
Programming	Python and C
Academic Service	Reviewer for <i>Mathematical Methods of Operations Research</i> , <i>Systems & Control Letters</i> , <i>Stochastic Analysis and Applications</i> , and <i>International Journal of Game Theory</i>

Professional Experience

Visiting Assistant Professor <i>Department of Mathematics, Indian Institute of Technology Dharwad, India</i> Teaching and research in applied probability, stochastic processes, mathematical finance, stochastic control, and stochastic games. Currently teaching Introduction to Mathematical Finance I (MA404T) and Stochastic Models (MA407T).	Sep 2026–Present
Guest Lecturer	Aug 2026

Department of Mathematics, Indian Institute of Technology Dharwad, India
Served as course instructor for Introduction to Mathematical Finance I (MA404T) and Stochastic Models (MA407T).
Continued teaching both courses after joining IIT Dharwad as Visiting Assistant Professor.

Postdoctoral Fellow Dec 2024–Sep 2026
Southern University of Science and Technology (SUSTech), Shenzhen, China
Supervisor: Prof. Jie Xiong. Research on singularly perturbed stochastic control, partially observable stochastic control, stochastic games, and stochastic differential games.

Visiting Researcher May 2026
Indian Institute of Technology Bhubaneswar, India
Invited for a one-month collaborative research visit with Prof. Avijit Dutt Banik.

Postdoctoral Fellow Jul 2023–Nov 2024
Indian Institute of Technology Bombay, Mumbai, India
Supervisor: Prof. K. Suresh Kumar. Research on singular perturbations in stochastic control and stochastic games.

Junior Research Fellow Jul 2017–Jul 2018
Indian Institute of Engineering Science and Technology, Shibpur, India
Research fellowship supported by CSIR-HRDG.

Academic Qualifications

Ph.D. in Mathematics Indian Institute of Technology Guwahati, India Coursework CPI: 8.52; supervisor: Dr. Chandan Pal; thesis: <i>Risk-sensitive Stochastic Control and Games</i> .	2023
M.Sc. in Applied Mathematics Indian Institute of Engineering Science and Technology, Shibpur, India 88%; CGPA: 9.3; special paper: Operations Research.	2016
B.Sc. in Mathematics (Honours) Narasinha Dutt College, University of Calcutta, India 74.875%.	2014
Higher Secondary Examination Khasmora High School, Howrah, India 83.4%.	2011
Secondary Examination Rudrapur Siksha Sadan, Howrah, India 78.62%.	2009
Diploma in Information Technology Applications Begri Youth Computer Training Centre (YCTC), Howrah, India One-year diploma, January–December 2011.	2011

Current and Proposed Research Direction

My immediate programme concerns risk-sensitive ergodic control of singularly perturbed continuous-time Markov decision processes, including finite-parameter optimality equations, aggregated limit models, asymptotically optimal policies, and numerical evaluation. Longer-term directions include partial-information control through belief-state methods, zero-sum and nonzero-sum mean-field games, and PDMDP/semi-Markov models. Applications include production–inventory and maintenance systems, queueing networks, mathematical finance, and reliable decision-making in cyber-physical systems.

Research Publications

Published and Accepted Journal Articles

- Subrata Golui** and Jie Xiong. “Continuous-time zero-sum games for Markov decision processes under the risk-sensitive first passage discounted cost criterion.” *Dynamic Games and Applications* (2025). DOI: 10.1007/s13235-025-00677-9. Citation: 0; Q2.
- Mrinal K. Ghosh, **Subrata Golui**, Chandan Pal, and Somnath Pradhan. “An eigenvalue approach to risk-sensitive nonzero-sum ergodic games for discrete-time Markov decision processes.” *Pure and Applied Functional Analysis* (2025), accepted. Citation: 0; Q4.
- Subrata Golui**. “Zero-sum games for piecewise deterministic Markov decision processes with risk-sensitive finite-horizon cost criterion.” *Stochastic Analysis and Applications*, 43, 326–352 (2025). DOI: 10.1080/07362994.2025.2470941.

Citation: 0; Q2.

4. **Subrata Golui**, Chandan Pal, Manikandan Rangaswamy, and Abhay Sobhanan. "Optimal control of a dynamic production-inventory system with various cost criteria." *Annals of Operations Research*, 337, 75–103 (2023). DOI: [10.1007/s10479-023-05716-5](https://doi.org/10.1007/s10479-023-05716-5). Citation: 3; Q1.
5. Mrinal K. Ghosh, **Subrata Golui**, Chandan Pal, and Somnath Pradhan. "Zero-sum games for continuous-time Markov decision processes with risk-sensitive average cost criterion." *Mathematical Control and Related Fields*, 14, 255–283 (2023). DOI: [10.3934/mcrf.2023003](https://doi.org/10.3934/mcrf.2023003). Citation: 2; Q2.
6. Mrinal K. Ghosh, **Subrata Golui**, Chandan Pal, and Somnath Pradhan. "Discrete-time zero-sum games for Markov chains with risk-sensitive average cost criterion." *Stochastic Processes and their Applications*, 158, 40–74 (2022). DOI: [10.1016/j.spa.2022.12.009](https://doi.org/10.1016/j.spa.2022.12.009). Citation: 11; Q1.
7. Mrinal K. Ghosh, **Subrata Golui**, Chandan Pal, and Somnath Pradhan. "Nonzero-sum risk-sensitive continuous-time stochastic games with ergodic costs." *Applied Mathematics and Optimization*, 86, 1–29 (2022). DOI: [10.1007/s00245-022-09878-9](https://doi.org/10.1007/s00245-022-09878-9). Citation: 7; Q1.
8. **Subrata Golui** and Chandan Pal. "Risk-sensitive discounted cost criterion for continuous-time Markov decision processes on a general state space." *Mathematical Methods of Operations Research*, 95, 219–247 (2022). DOI: [10.1007/s00186-022-00779-9](https://doi.org/10.1007/s00186-022-00779-9). Citation: 6; Q2.
9. **Subrata Golui**, Chandan Pal, and Subhamay Saha. "Continuous-time zero-sum games for Markov decision processes with discounted risk-sensitive cost criterion." *Dynamic Games and Applications*, 12, 485–512 (2021). DOI: [10.1007/s13235-021-00391-2](https://doi.org/10.1007/s13235-021-00391-2). Citation: 8; Q2.
10. **Subrata Golui** and Chandan Pal. "Continuous-time zero-sum games for Markov chains with risk-sensitive finite-horizon cost criterion." *Stochastic Analysis and Applications*, 40, 78–95 (2021). DOI: [10.1080/07362994.2021.1889381](https://doi.org/10.1080/07362994.2021.1889381). Citation: 12; Q2.
11. **Subrata Golui** and Chandan Pal. "Continuous-time zero-sum games for Markov decision processes with discounted risk-sensitive cost criterion on a general state space." *Stochastic Analysis and Applications*, 41, 327–357 (2021). DOI: [10.1080/07362994.2021.2013889](https://doi.org/10.1080/07362994.2021.2013889). Citation: 7; Q2.

Manuscripts Under Review

12. **Subrata Golui** and Jie Xiong. "Nonzero-sum risk-sensitive stochastic games for continuous-time Markov chains with discounted cost criterion." Under review.
13. **Subrata Golui**, Chandan Pal, Somnath Pradhan, and Jie Xiong. "Zero-sum games under the risk-sensitive discounted cost criterion for piecewise deterministic Markov decision processes." Under review.
14. **Subrata Golui** and Jie Xiong. "Discounted zero-sum games for singularly perturbed discrete-time Markov chains." Under review.

Fellowships, Qualifications and Honours

- SUSTech Postdoctoral Fellowship, supported by Shenzhen Municipality and Guangdong Province (2024–2026).
- IIT Bombay Institute Postdoctoral Fellowship (2023–2024).
- IIT Guwahati iPhD Fellowship, Ministry of Human Resource Development (2018–2023).
- NBHM Ph.D. Scholarship (2019–2020).
- CSIR Junior Research Fellowship, IEST Shibpur (2017–2018).
- Qualified CSIR-UGC NET in Mathematics five times, including four qualifications with JRF (2016–2018).
- GATE in Mathematics (2017).
- West Bengal Government Merit-cum-Means Scholarship during B.Sc. (2012–2014) and M.Sc. studies.

Teaching Experience

Course Instructor

- **Introduction to Mathematical Finance I (MA404T)**, IIT Dharwad – Course Instructor (Aug 2026–Present).
- **Stochastic Models (MA407T)**, IIT Dharwad – Course Instructor (Aug 2026–Present).

Teaching Assistant

- **Calculus II**, SUSTech, China – undergraduate teaching assistantship (2025).
- **Probability I**, IIT Bombay – M.Sc. teaching assistantship (2023 and 2024).
- **Derivative Pricing**, IIT Bombay – undergraduate teaching assistantship (2024).
- **Real Analysis**, IIT Guwahati – undergraduate teaching assistantship (2023).
- **Discrete-Time Markov Chain and Poisson Processes**, NPTEL, IIT Guwahati (2022).
- **Probability Theory and Random Processes**, IIT Guwahati – undergraduate, M.Sc., and Ph.D. teaching assistantships (2019–2021).

Selected Academic Engagements

- Seminar talk, Department of Mathematics, IIT Bhubaneswar, India (2026).
- Keynote talk, Mathematics Seminar, McGill University, Montreal, Canada (2024).
- Invited talk, Workshop on Stochastic Control and Games, Alfréd Rényi Institute of Mathematics, Budapest, Hungary (29–31 January 2024).
- Contributed talk, International Conference on Advances in Applied Probability and Stochastic Processes, St. Aloysius College, Thrissur, India (18–20 January 2024).
- Participant, Probability and Stochastic Processes, ICTS Bengaluru, India (17–21 September 2023).
- Participant, Winter School on Graphs and Stochastic Processes, IISER Pune, India (16–20 December 2019).
- Participant, Summer Outreach Program in Mathematics, NISER Bhubaneswar, India (20 June–2 July 2016).

Research Collaboration and Professional Service

Collaborative work has been undertaken with Dr. Chandan Pal, Dr. Subhamay Saha, Prof. Mrinal K. Ghosh, Dr. Somnath Pradhan, Dr. Manikandan Rangaswamy, Dr. Abhay Sobhanan, Prof. Jie Xiong, and Prof. Avijit Dutt Banik.

I have contributed to peer review for *Mathematical Methods of Operations Research*, *Systems & Control Letters*, *Stochastic Analysis and Applications*, and the *International Journal of Game Theory*.

Personal Information and Academic References

Personal Information

Nationality	Indian
Date of Birth	21 June 1993
Father's Name	Gobordhan Golui
Mother's Name	Radha Golui
Present Address	Department of Mathematics, Room 506, A2 Building, Indian Institute of Technology Dharwad, Dharwad, Karnataka, India – 580011
Permanent Address	Village and P.O. Rudrapur, P.S. Domjur, District Howrah, West Bengal, India – 711411

Academic References

Dr. Chandan Pal	Associate Professor, IIT Guwahati; Ph.D. supervisor; cpal@iitg.ac.in ; +91 361 258 2640
Prof. K. Suresh Kumar	Professor, IIT Bombay; former postdoctoral mentor; suresh@math.iitb.ac.in ; +91 22 2576 7489
Prof. Mrinal K. Ghosh	Retired Professor, IISc Bengaluru; collaborator; mkg@math.iisc.ernet.in
Dr. Subhamay Saha	Associate Professor, IIT Guwahati; collaborator; saha.subhamay@iitg.ac.in ; +91 361 258 2638
Prof. Jie Xiong	Chair Professor, SUSTech, China; former postdoctoral supervisor and collaborator; xiongj@sustech.edu.cn